JIAF Haiti 2025: Deprivation Index Sensitivity Analysis

Analysis Date: January 21, 2026
Communes Analyzed: 140

This sensitivity analysis examines how methodological choices affect deprivation index estimates across Haiti's 146 communes. We compare six index variants across two dimensions:

1. AGGREGATION APPROACH

• Compensatory: Arithmetic mean allows high scores in one sector to offset low scores in others
• Max−Dominant: Takes maximum across sectors, no compensation allowed

2. STANDARDIZATION METHOD

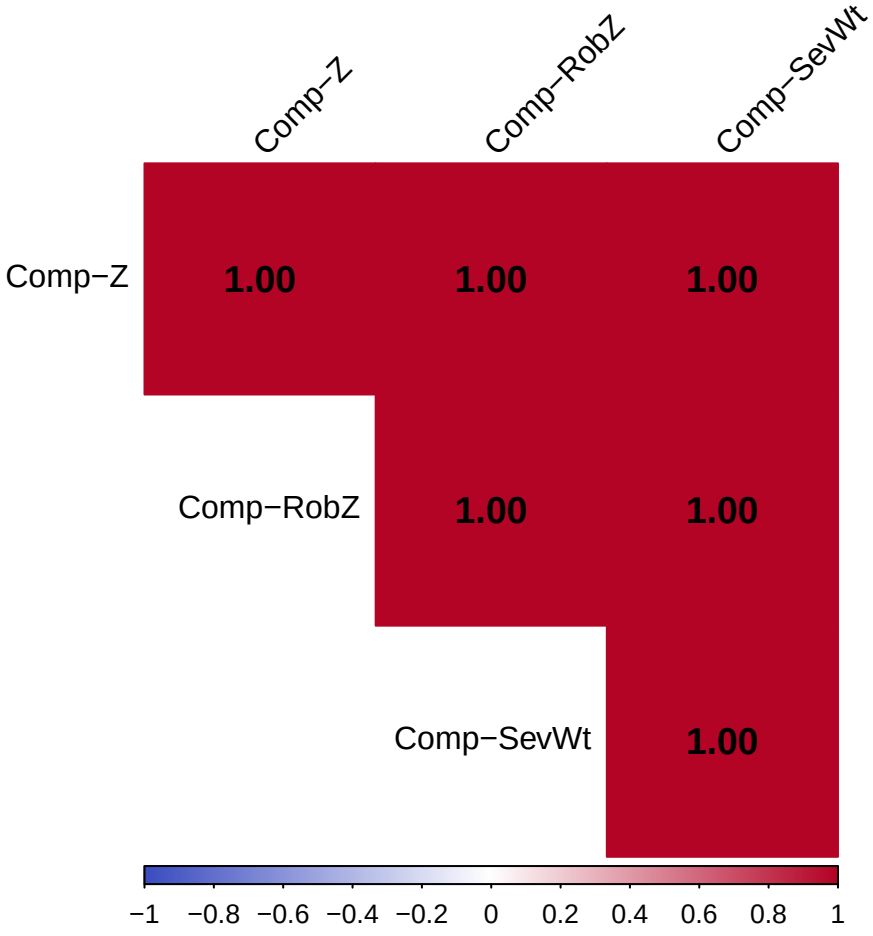
• Z−score: Standard deviation−based standardization (mean = 0, SD = 1)
• Robust Z: Median−based standardization (resistant to outliers)
• Severity−weighted: JIAF humanitarian severity levels (1−5 scale)

Key Questions:

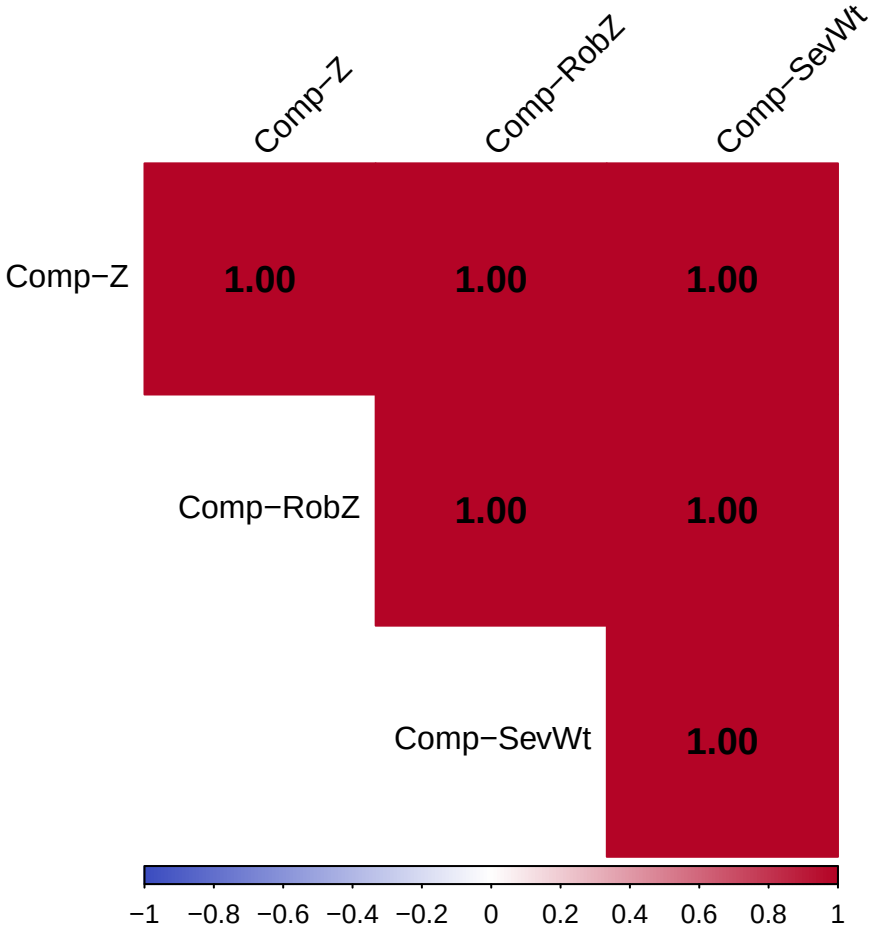
• Does severity−weighting substantially change spatial patterns compared to statistical standardization?
• Are results robust to aggregation approach, or do we need to report multiple indices?
• Which communes are consistently high/low across methods (robust findings)?
• Where does methodological choice matter most (uncertainty hotspots)?

Section 1: Standardization Method Sensitivity − Compensatory Approach

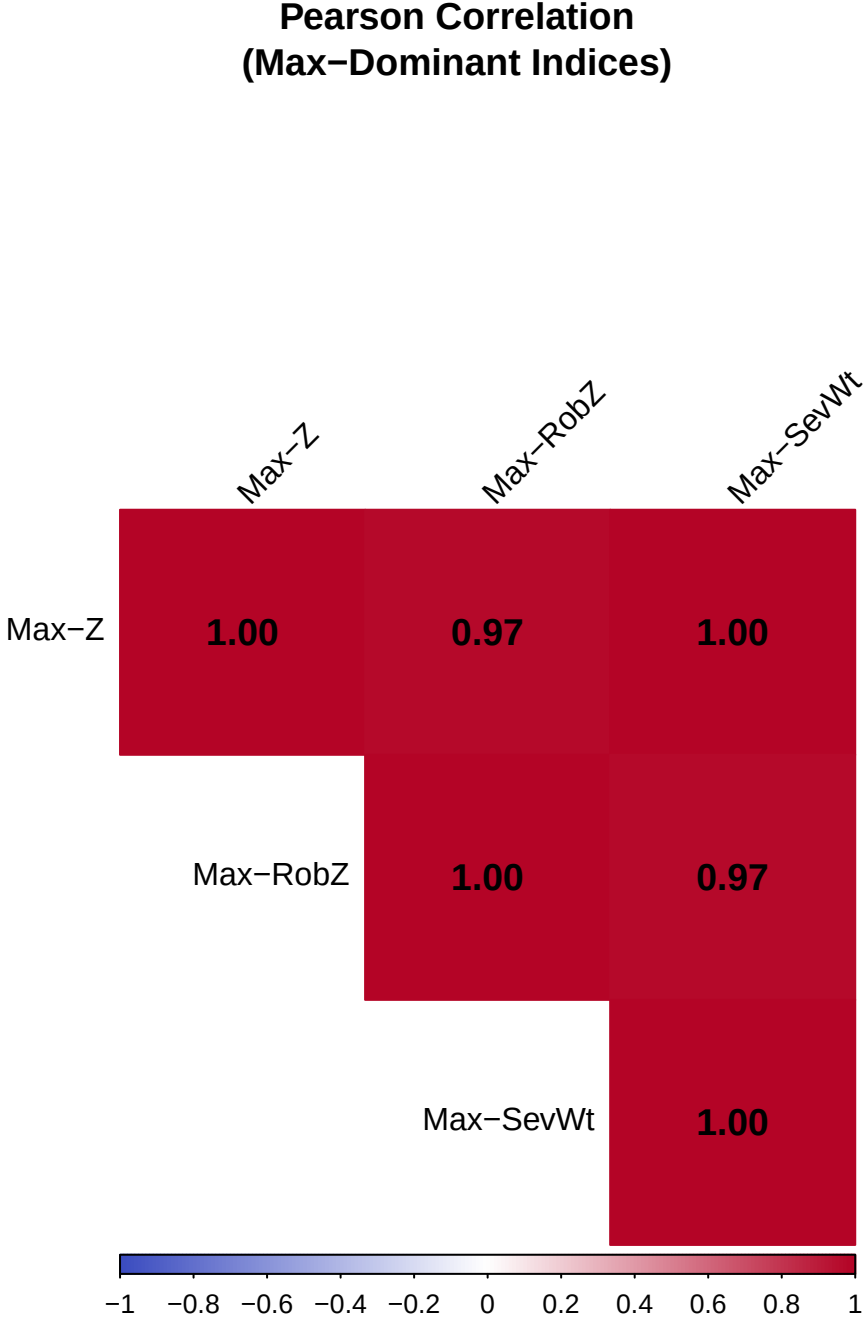
Pearson Correlation
(Compensatory Indices)

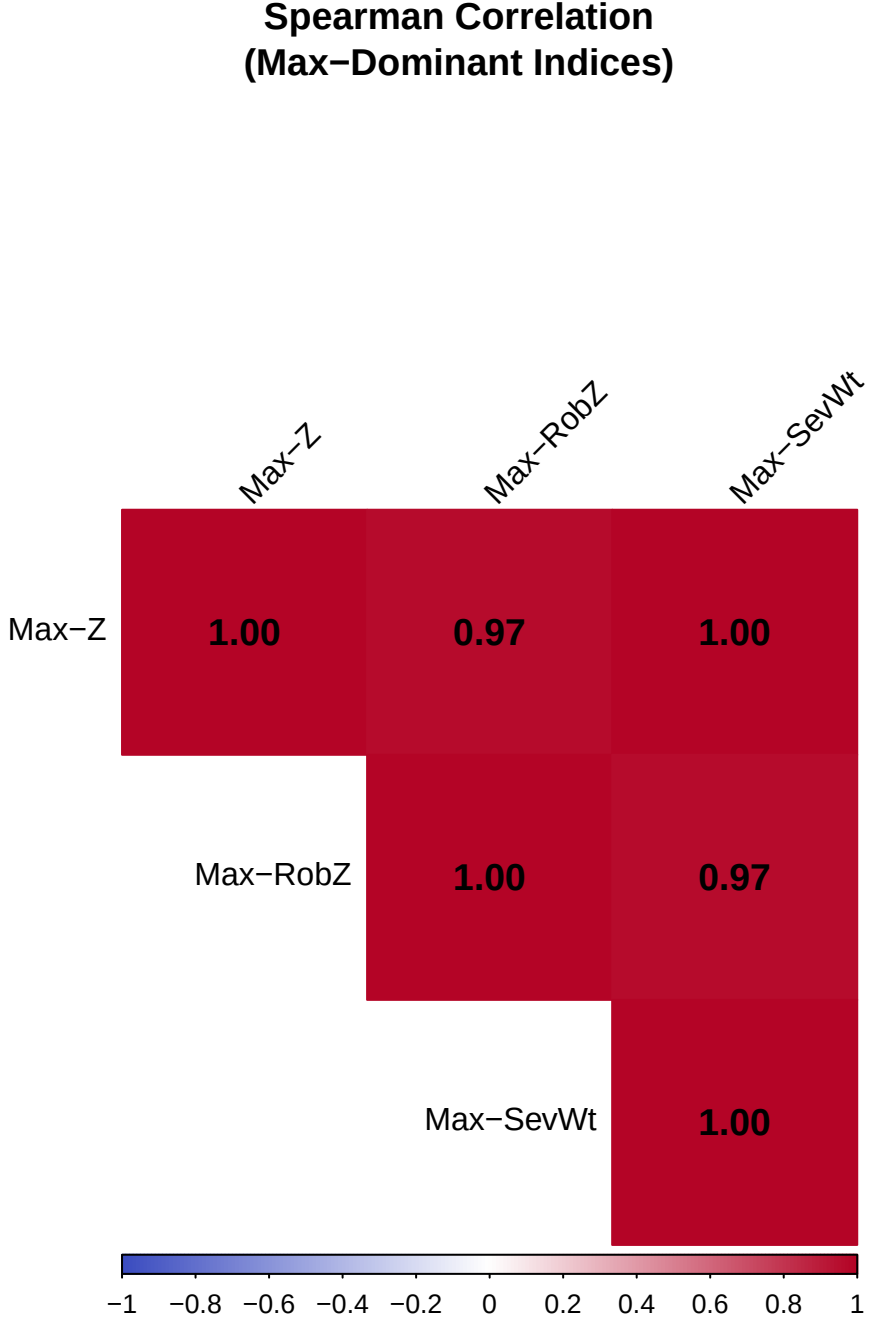


Spearman Correlation
(Compensatory Indices)



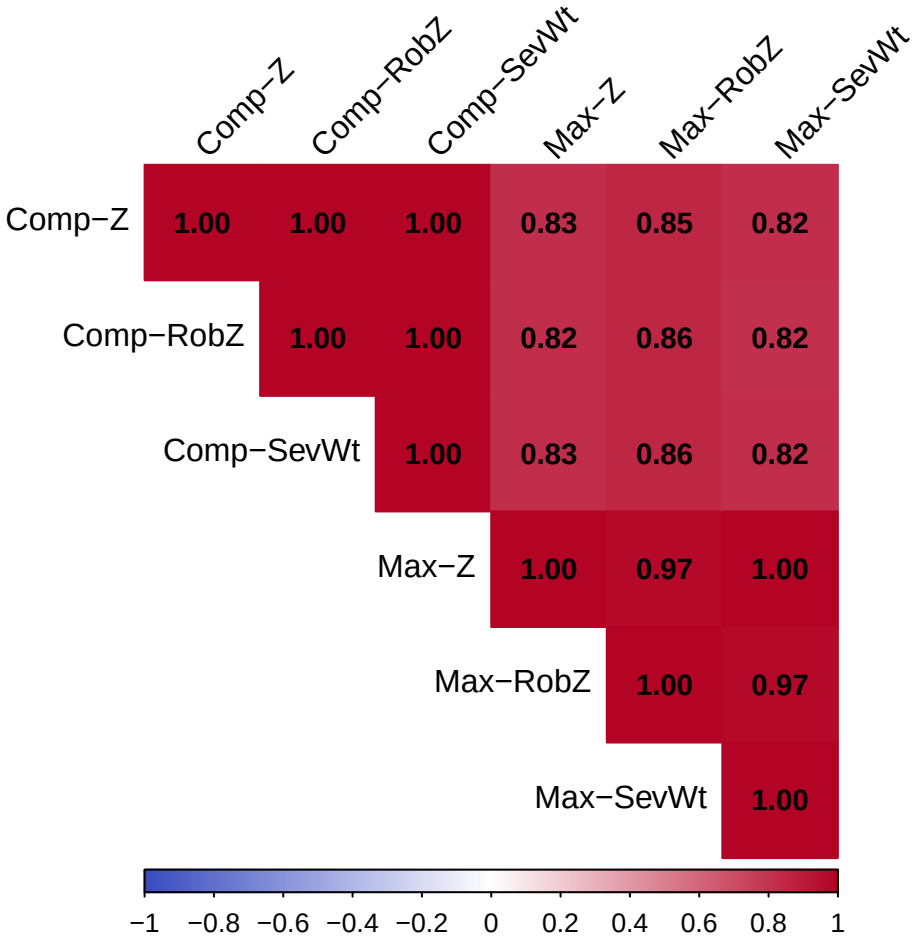
Section 1 (continued): Standardization Method Sensitivity − Max−Dominant Approach


Pearson Correlation
(Max−Dominant Indices)
Max−Z
Max−RobZ
Max−SevWt
1.00
0.97
1.00
1.00
0.97
1.00
−1 −0.8 −0.6 −0.4 −0.2 0 0.2 0.4 0.6 0.8 1

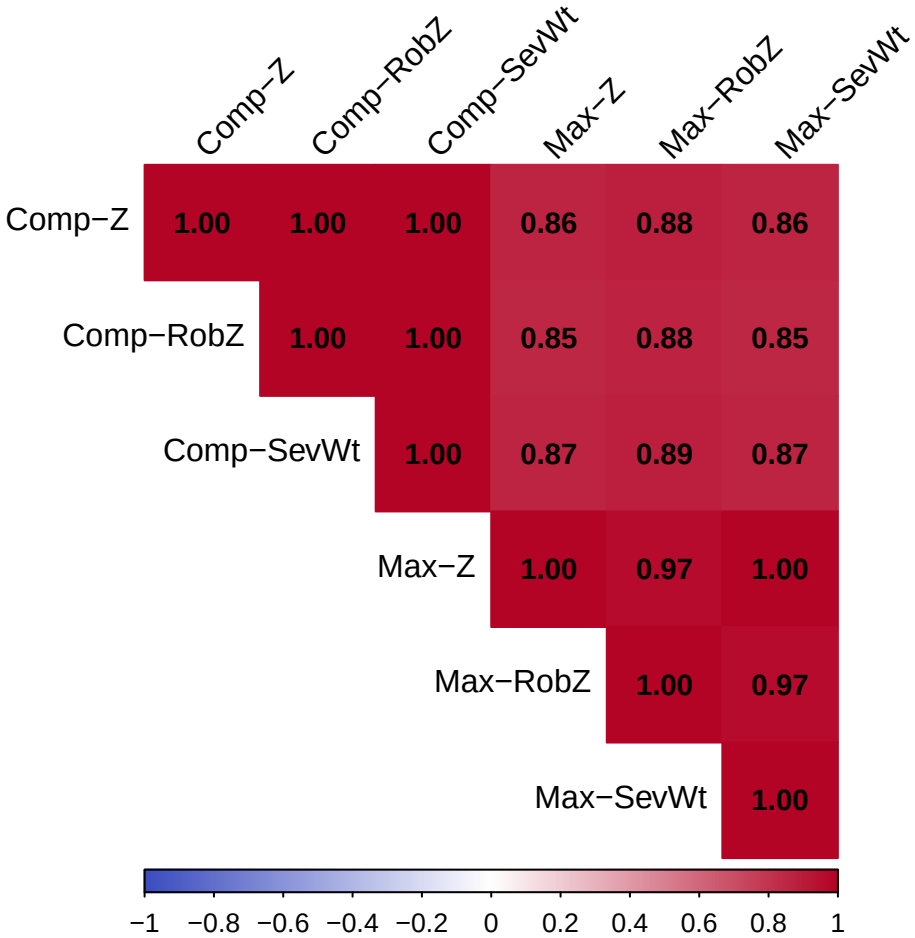

Spearman Correlation
(Max−Dominant Indices)
Max−Z
Max−RobZ
Max−SevWt
1.00
0.97
1.00
1.00
0.97
1.00
−1 −0.8 −0.6 −0.4 −0.2 0 0.2 0.4 0.6 0.8 1

Section 1 (continued): Cross−Method Correlation Matrix

Pearson Correlation (All Six Indices)



Spearman Correlation (All Six Indices)



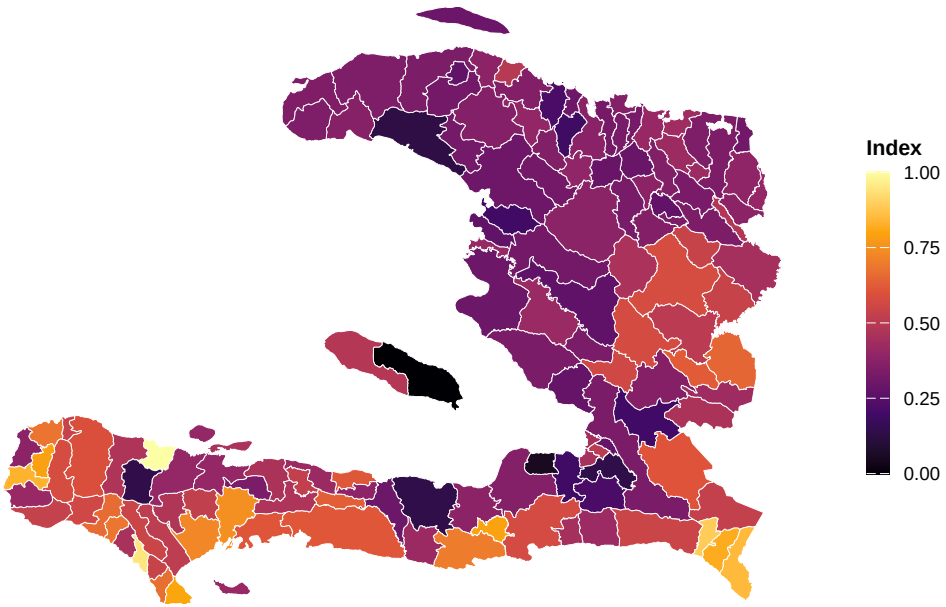
Max−Dominant: r = 0.98 | between approaches: r = 0.834

Section 2: Spatial Comparison − Compensatory Approach

How does standardization method affect spatial patterns?

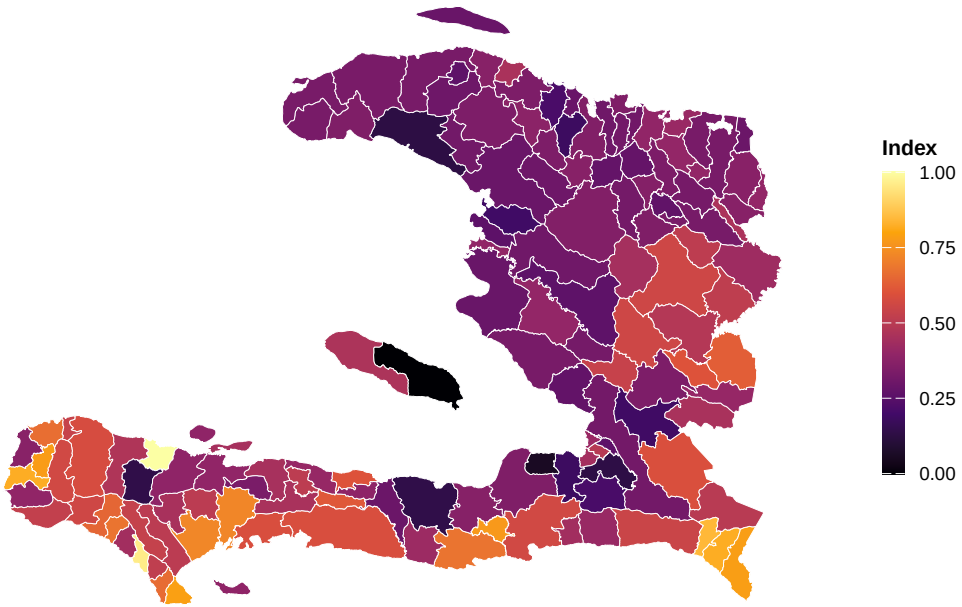
A. Compensatory − Z−score

Standard deviation standardization



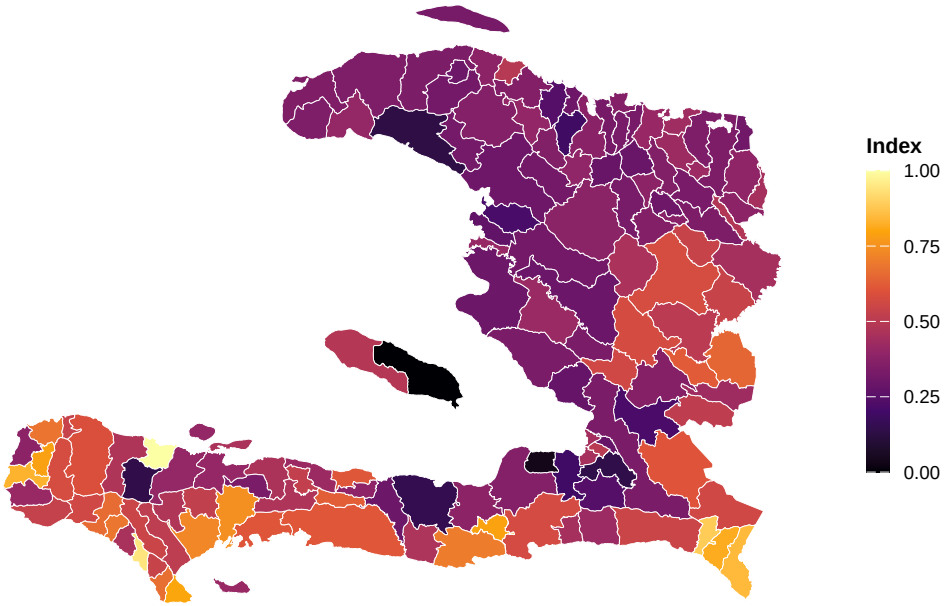
B. Compensatory − Robust Z

Median−based standardization



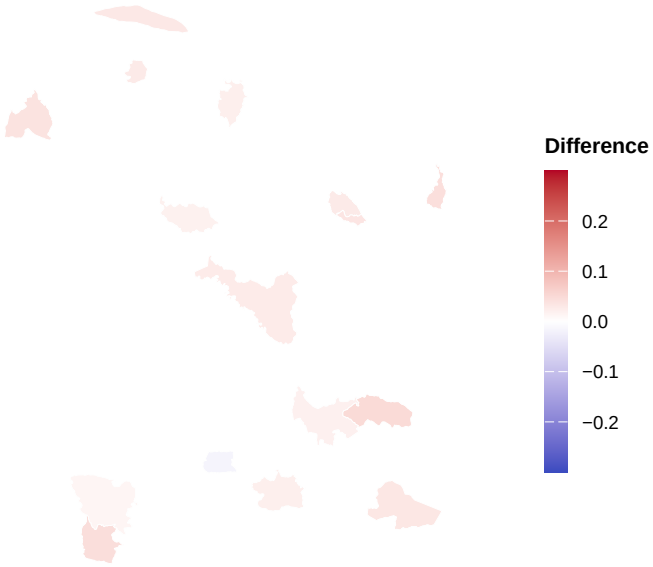
C. Compensatory − Severity−Weighted

JIAF humanitarian severity levels



D. Difference: Sev−Wt minus Z−score

Positive = higher with severity weighting

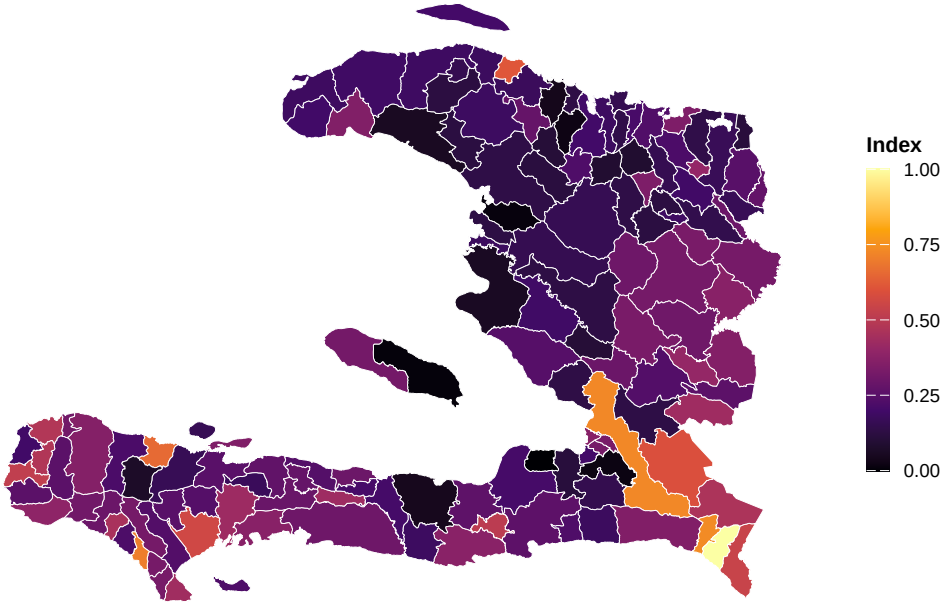


Section 2 (continued): Spatial Comparison − Max−Dominant Approach

How does standardization method affect spatial patterns?

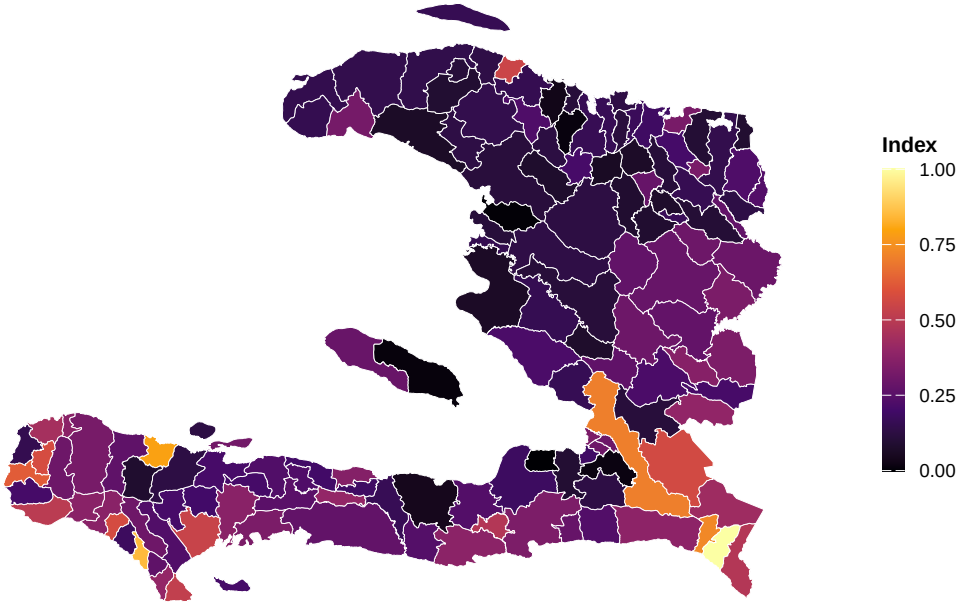
A. Max−Dominant − Z−score

Standard deviation standardization



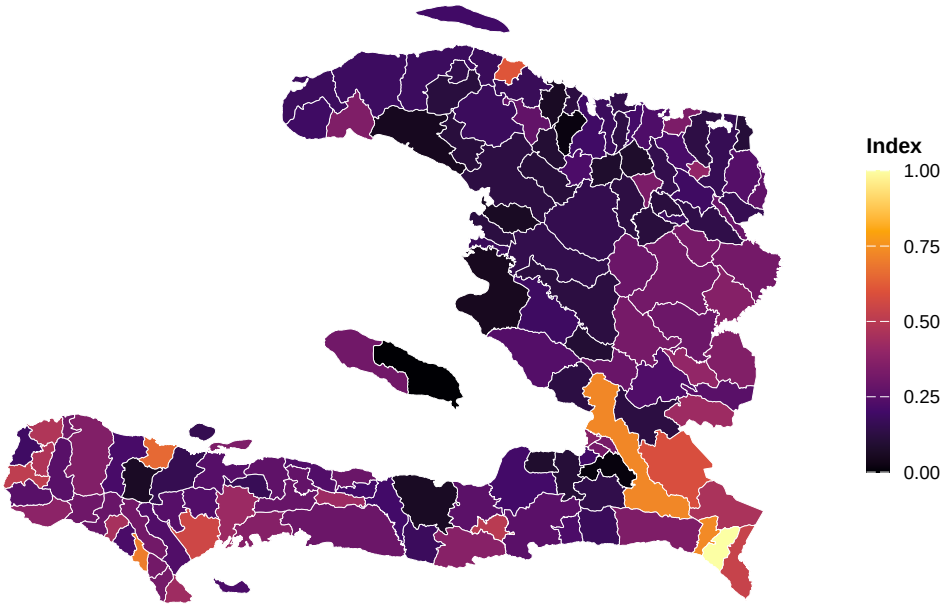
B. Max−Dominant − Robust Z

Median−based standardization



C. Max−Dominant − Severity−Weighted

JIAF humanitarian severity levels



D. Difference: Sev−Wt minus Z−score

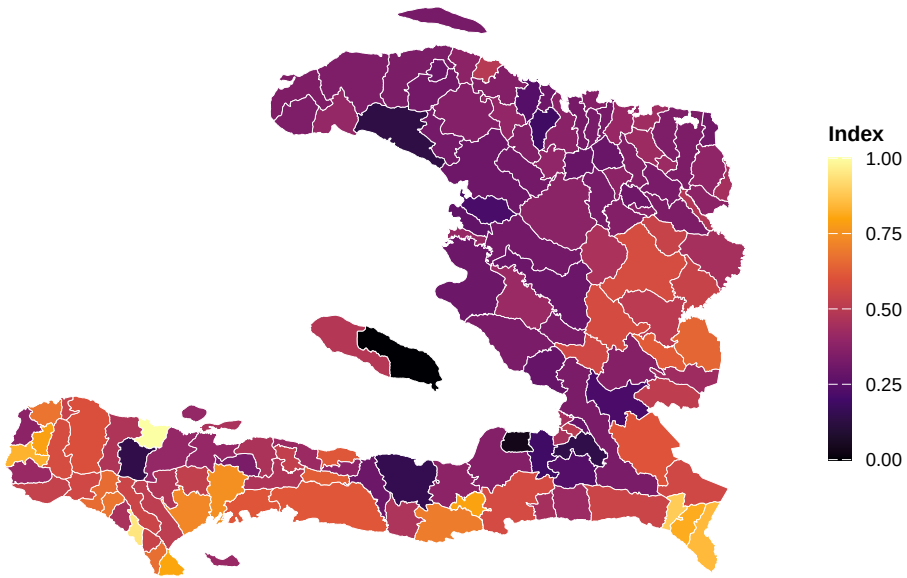
Positive = higher with severity weighting

Section 3: Aggregation Approach Comparison

Compensatory vs. Max−Dominant (using severity−weighted standardization)

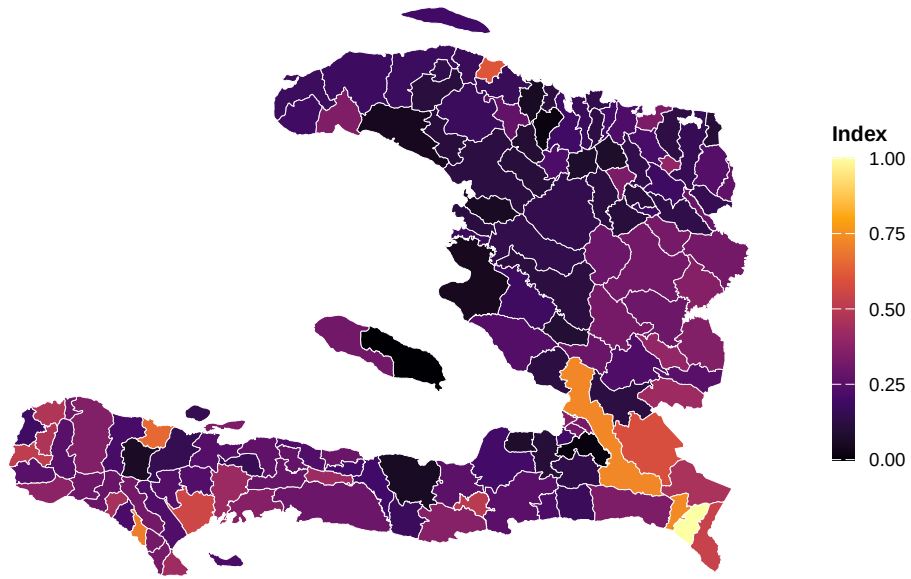
A. Compensatory (Sev−Wt)

Arithmetic mean of severity−weighted sectors



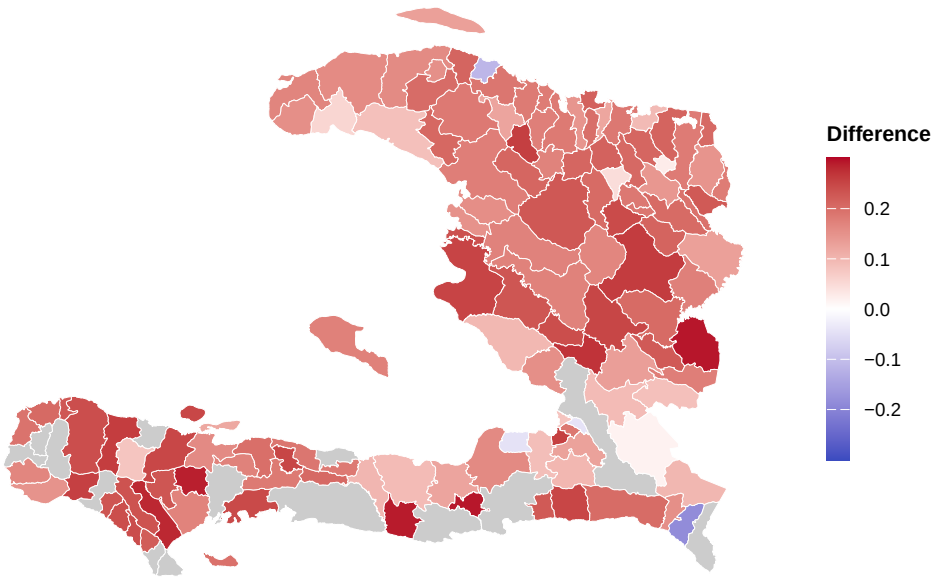
B. Max−Dominant (Sev−Wt)

Maximum severity−weighted sector



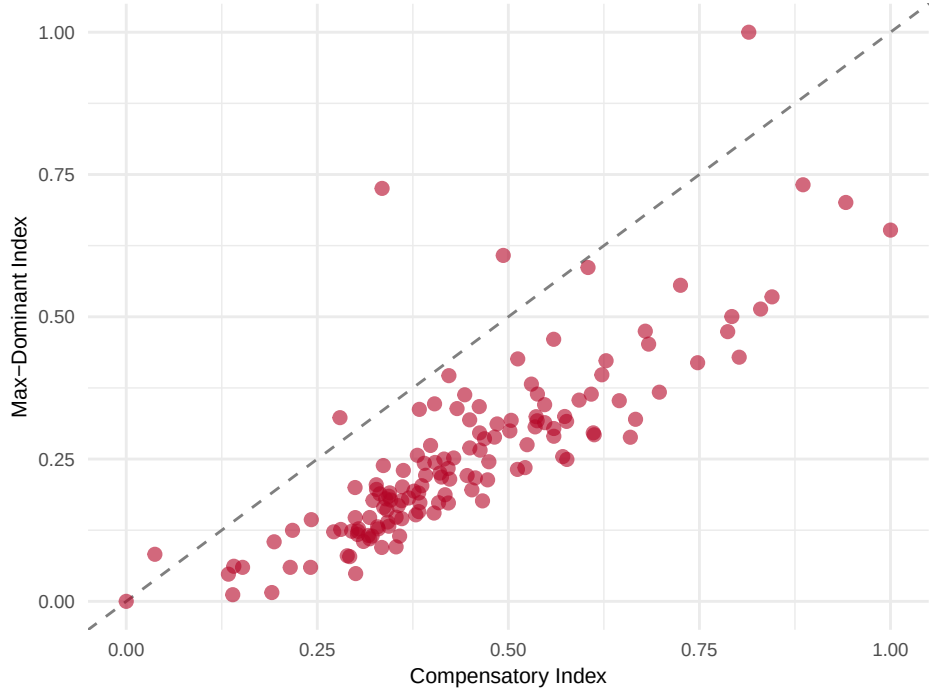
C. Difference: Comp minus Max

Positive = higher with compensatory approach



D. Compensatory vs Max−Dominant

r = 0.823



Section 4: Rank−Order Stability Analysis

Top 10 Commune Overlap

	Comp−Z	Comp−RobZ	Comp−SevWt	Max−Z	Max−RobZ	Max−SevWt
Max−SevWt	6	7	6	10	8	10
Max−RobZ	6	6	6	8	10	8
Max−Z	6	7	6	10	8	10
Comp−SevWt	10	9	10	6	6	6
Comp−RobZ	9	10	9	7	6	7
Comp−Z	10	9	10	6	6	6

Mean Absolute Rank Difference

	Comp−Z	Comp−RobZ	Comp−SevWt	Max−Z	Max−RobZ	Max−SevWt
Max−SevWt	14.8	15.1	14.3	0.2	7.1	0
Max−RobZ	14.2	14	13.6	6.9	0	7.1
Max−Z	14.8	15.1	14.3	0	6.9	0.2
Comp−SevWt	1.5	2.1	0	14.3	13.6	14.3
Comp−RobZ	1.7	0	2.1	15.1	14	15.1
Comp−Z	0	1.7	1.5	14.8	14.2	14.8

Section 4 (continued): Top 10 Most Deprived Communes by Index

Rank	Commune	Department	Index
1	Corail	Grande'Anse	1.000
2	Roche a Bateau	South	0.942
3	Grand Gosier	South−East	0.886
4	Anse−a−Pitre	South−East	0.845
5	Anse d'Hainault	Grande'Anse	0.830
6	Thiotte	South−East	0.815
7	Saint Jean du Sud	South	0.802
8	La Vallee	South−East	0.792
9	Chambellan	Grande'Anse	0.787
10	Cavaillon	South	0.748

Rank	Commune	Department	Index
1	Corail	Grande'Anse	1.000
2	Roche a Bateau	South	0.958
3	Grand Gosier	South−East	0.842
4	Anse d'Hainault	Grande'Anse	0.819
5	Thiotte	South−East	0.818
6	Saint Jean du Sud	South	0.787
7	Anse−a−Pitre	South−East	0.781
8	Chambellan	Grande'Anse	0.778
9	La Vallee	South−East	0.769
10	Les Cayes	South	0.724

Rank	Commune	Department	Index
1	Corail	Grande'Anse	1.000
2	Roche a Bateau	South	0.942
3	Grand Gosier	South−East	0.886
4	Anse−a−Pitre	South−East	0.845
5	Anse d'Hainault	Grande'Anse	0.830
6	Thiotte	South−East	0.815
7	Saint Jean du Sud	South	0.802
8	La Vallee	South−East	0.792
9	Chambellan	Grande'Anse	0.787
10	Cavaillon	South	0.748

Rank	Commune	Department	Index
1	Thiotte	South−East	1.000
2	Grand Gosier	South−East	0.734
3	Croix−Des−Bouquets	West	0.728
4	Roche a Bateau	South	0.703
5	Corail	Grande'Anse	0.655
6	Anse−a−Foleur	North−West	0.611
7	Ganthier	West	0.590
8	Les Cayes	South	0.559
9	Anse−a−Pitre	South−East	0.539
10	Anse d'Hainault	Grande'Anse	0.518

Rank	Commune	Department	Index
1	Thiotte	South−East	1.000
2	Roche a Bateau	South	0.847
3	Corail	Grande'Anse	0.789
4	Grand Gosier	South−East	0.728
5	Croix−Des−Bouquets	West	0.704
6	Anse d'Hainault	Grande'Anse	0.623
7	Chambellan	Grande'Anse	0.576
8	Port−a−Piment	South	0.573
9	Ganthier	West	0.567
10	Anse−a−Foleur	North−West	0.547

Rank	Commune	Department	Index
1	Thiotte	South−East	1.000
2	Grand Gosier	South−East	0.732
3	Croix−Des−Bouquets	West	0.726
4	Roche a Bateau	South	0.701
5	Corail	Grande'Anse	0.652
6	Anse−a−Foleur	North−West	0.608
7	Ganthier	West	0.587
8	Les Cayes	South	0.555
9	Anse−a−Pitre	South−East	0.535
10	Anse d'Hainault	Grande'Anse	0.514

Section 5: Summary Statistics

Index	Mean	SD	Min	Q25	Median	Q75	Max
Comp−Z	0.438	0.174	0	0.334	0.408	0.535	1
Comp−RobZ	0.425	0.172	0	0.320	0.394	0.515	1
Comp−SevWt	0.441	0.173	0	0.334	0.410	0.537	1
Max−Z	0.263	0.159	0	0.155	0.235	0.329	1
Max−RobZ	0.250	0.173	0	0.132	0.211	0.326	1
Max−SevWt	0.258	0.159	0	0.148	0.228	0.323	1

Note: All indices normalized to 0−1 scale. Higher values indicate greater deprivation.
Severity−weighted indices use JIAF humanitarian severity levels (1−5).
Z−score and Robust Z standardize using statistical moments.

Key Findings and Interpretation

1. STANDARDIZATION METHOD SENSITIVITY

Compensatory Approach:
- Correlation between Z−score and Severity−weighted: r = 0.998
- Top 10 overlap: 10/10 communes
- Interpretation: Very high agreement

Max−Dominant Approach:
- Correlation between Z−score and Severity−weighted: r = 0.998
- Top 10 overlap: 10/10 communes
- Interpretation: Very high agreement

2. AGGREGATION APPROACH SENSITIVITY

- Correlation (Compensatory vs Max−Dominant, Sev−Wt): r = 0.823
- Top 10 overlap: 6/10 communes
- Interpretation: Aggregation approach substantially affects rankings

3. ROBUSTNESS ASSESSMENT

- Mean correlation within Compensatory indices: r = 0.998
- Mean correlation within Max−Dominant indices: r = 0.980
- Mean correlation between approaches: r = 0.834
- Implication: Standardization method has minimal impact within approaches

4. RECOMMENDATIONS

- Multiple indices recommended due to approach sensitivity
- Report both Compensatory and Max−Dominant
- Severity−weighting provides humanitarian policy relevance